

FGI & FCS

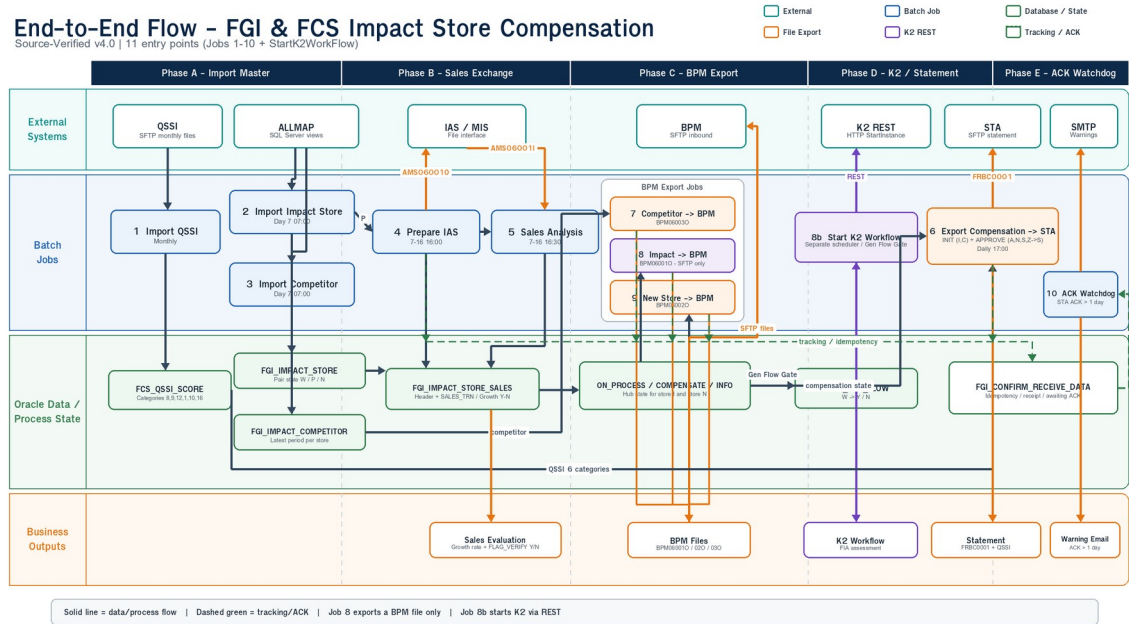
Batch Job Technical Document

Store Business Partner - Franchise Growth Impact / Compensation

Version 4.0 - Source-Verified Implementation & Operations Edition

End-to-End Flow - FGI & FCS Impact Store Compensation

Source-Verified v4.0 | 11 entry points (Jobs 1-10 + StartK2Workflow)



Document Version	4.0 - updated from Improved v3.0 using SGI V4
Scope	11 entry points: Jobs 1-10 + StartK2Workflow; logic, operations and data dictionary
Primary DB	Oracle fcs_frn; external SQL Server GSMALLMAP / allmapssa
Source of truth	main -> controller -> service -> DAO -> SQL + config.xml
Classification / Updated	Internal Use Only / 1 July 2026

1. Executive Summary

Purpose เอกสารฉบับนี้อัปเดตจาก Version 3.0 โดยใช้ SGI V4 เป็น source of truth เพื่อให้ใช้ทั้งการปฏิบัติการ, troubleshooting และ re-implement ระบบเป็นภาษาอื่นได้โดยไม่ตีความ business logic ผิด

1.1 Scope และหลักการใช้งาน

- ครอบคลุม 10 batch jobs เดิม และ entry point เพิ่มเติม Job 8b - StartK2Workflow ซึ่งเป็นตัวเรียก K2 REST จริง
- โครงแบบ System Design & Operation Guide ของ v3.0 พร้อมเพิ่ม implementation-grade specification, Data Dictionary, status domains, sequences และ gotchas
- ข้อมูลที่ SGI V4 ระบุว่าเป็น current-state bug หรือความเสี่ยง ถูกบันทึกแยกจากแนวทางปรับปรุง เพื่อไม่ให้เผลอเปลี่ยนพฤติกรรมระหว่าง re-implement
- ข้อมูล credential แสดงแบบ masked; เอกสารระบุเฉพาะตำแหน่งจัดเก็บและความเสี่ยงเท่านั้น

1.2 Major Changes from v3.0

Area	Update
Architecture	แยก Job 8 (BPM06001O ผ่าน SFTP) ออกจาก Job 8b (K2 REST StartInstance) อย่างชัดเจน
Job 2	แก้ branch type set ให้รวม B และ FPT1; คู่มือเดิมถูก skip เพราะ updateList เป็น dead code
Job 5	เพิ่ม pre-accept, raw TOTAL_WORKING_DAY, paired outlier logic และ NULL growth diff -> auto accept Y
Job 6	เพิ่ม dynamic eligibility, QSSI completeness gate, 10 ordered DB mutations, Thai/Buddhist field formatting และ known purge bug
Job 1/4/7	เพิ่ม off-by-one month, transaction/auto-commit, tracking semantics และ gotchas ที่มีผลต่อ re-run
Data Model	เพิ่ม hub relationship, polymorphic tracking key, status domains, 7 sequence catalog และ external views
Cross-cutting	แก้ข้อเท็จจริงด้าน plaintext secrets, STQA datasource, mail routing และ transaction pattern ต่อ job

2. Source-Verified Errata from v3.0

Important ตารางนี้เป็นรายการที่หากยึด v3.0 เดิมเพียงอย่างเดียว อาจทำให้ implementation หรือ operational decision ผิด

ID	Level	Job	v3.0 assumption	Source-verified truth	Impact
E1	CRITICAL	8/8b	Job 8 รวม SFTP + K2	Job 8 ทำ SFTP เท่านั้น; StartK2Workflow เป็น entry point แยก	สถาปัตยกรรมและ recovery ผิด
E2	CRITICAL	8b	Gen Flow Gate อยู่ใน Job 8	Gate อยู่ใน StartK2Workflow	ชุด file export กับ workflow start ถูกผูกผิด
E3	CRITICAL	2	Eligible มี 6 branch types	กฎจริงรวม B และ FPT1 พร้อมเงื่อนไข	คัดค้านผิด

ID	Level	Job	v3.0 assumption	Source-verified truth	Impact
				cancel type 06	
E4	CRITICAL	2	มี update existing pair	updateList เป็น dead code; existing pair ถูก skip	พฤติกรรมต่างจากรบบจริง
E5	CRITICAL	5	ตัด outlier อย่างเดียว	มี paired past/present flags; NULL diff > NVL(-1) -> Y	ผล Y/N ผิด
E6	CRITICAL	6	Sync DB เป็นขั้นตอนเดียว	มี 10 ordered mutations	state transition ไม่ครบ
E7	HIGH	6	Status filter คงที่	Dynamic A,N,S + optional I,C + stale Z	ส่งผิดชุด
E8	HIGH	6	QSSI เป็น optional	INIT ต้องครบ 6 categories; ใช้ max period เดียว	สร้าง INIT ผิด
E9	HIGH	6	Date formatting ทั่วไป	Field 3/5/6 ใช้ Thai/Buddhist locale; Z->S output only	ไฟล์ผิดสัญญา
E10	HIGH	1	ชื่อไฟล์และ period เดียวกัน	Filename เดือนปัจจุบัน แต่ DB period เดือนก่อนหน้า	off-by-one โดยเจตนา
E11	HIGH	1	SFTP ทั่วไป	Port 218; backup แม้ fail; numeric/string literal quirks	retry และ type mismatch
E12	HIGH	4	มี transaction	ไม่มี transaction; W->P commit ก่อนเขียนไฟล์	DB/file inconsistency
E13	HIGH	5	working day = filtered days	TOTAL_WORKING_DAY เป็น raw count; pre-accept ก่อน growth	flag ผิด
E14	HIGH	7	ใช้ค่า source ตรง	create_by=system; period จาก on-process; incomplete manager ถูก drop	ข้อมูลไฟล์ต่าง
E15	HIGH	3	upsert ได้	idempotency ระดับ period; ต้อง delete ก่อน re-import	duplicate/recovery
E16	MEDIUM	Security	ใช้ secret store	Secrets อยู่ plaintext/hardcoded ใน current state	security assumption ผิด
E17	MEDIUM	Environment	เป็น production	Properties ที่ commit ที่ STQA test	รันผิด environment
E18	MEDIUM	Sequences	ไม่ได้ระบุ	ต้องใช้ sequence catalog	PK generation ขาด
E19	MEDIUM	Mail	routing เดียว	ผู้รับต่างกันตาม job + IS_TEST switch	แจ้งผิดคน
E20	MEDIUM	6	purge tracking ปกติ	SQL ใช้ data_name string เดียวผิดรูปแบบ จึงไม่ purge	ข้อมูลสะสม

3. System Context and End-to-End Flow

3.1 System Context

- QSSI SFTP provides monthly score files; Job 1 maps them into FCS_QSSI_SCORE.
- ALLMAP SQL Server provides impact-store pairs and competitor records for Jobs 2 and 3.
- IAS/MIS exchanges AMS06001O outbound and AMS06001I inbound sales files for Jobs 4 and 5.
- BPM receives three independent SFTP files from Jobs 7-9; K2 workflow is started separately by Job 8b.
- STA receives FRBC0001 from Job 6; Job 10 monitors overdue ACK only for STA compensate datasets.

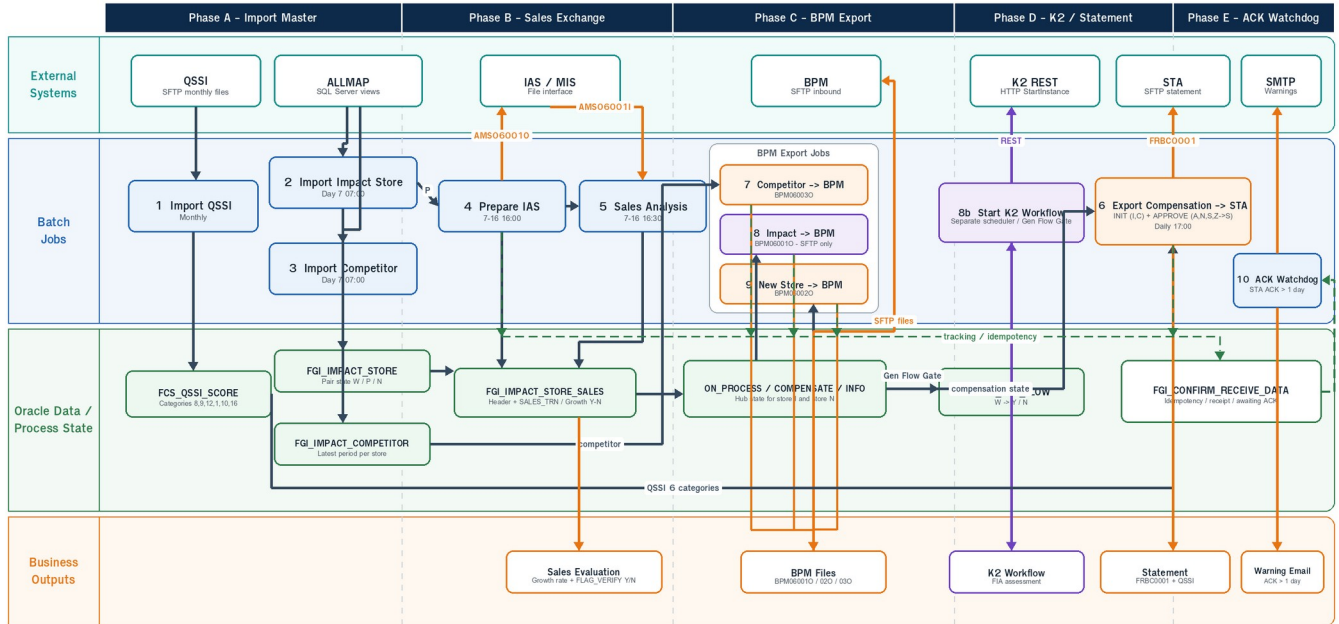
End-to-End Flow - Source-Verified

v4.0

11 ENTRY POINTS

End-to-End Flow – FGI & FCS Impact Store Compensation

Source-Verified v4.0 | 11 entry points (Jobs 1-10 + StartK2Workflow)



Job 8 exports BPM file only; Job 8b starts K2 separately; Job 10 monitors STA ACK records.

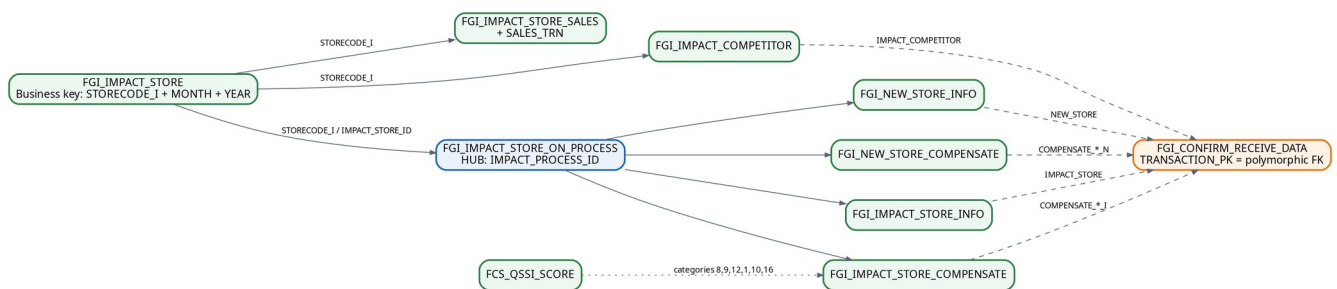
3.2 Entry Points and Schedule

#	Main Class	Script	Cron	Role
1	fcs.main.ImportQSSI	FCS_ImportQSSI.sh	Monthly	QSSI SFTP -> FCS_QSSI_SCORE
2	fgi.main.ImportImpactStore	FGI_ImportImpactStore.sh	0 07 7 * *	ALLMAP impact pairs -> W/P/N
3	fgi.main.ImportImpactCompetitor	Verify scheduler script	0 07 7 * *	ALLMAP competitors -> period import
4	fgi.main.PrepareImpactStoreToIAS	FGI_ExportImpactStoreToAMS.sh	0 16 7-16 * *	AMS060010 outbound
5	fgi.main.ImportImpactSaleFromIAS	FGI_ImportImpactStoreSale.sh	30 16 7-16 * *	AMS06001I -> growth/flag
6	fgi.main.ExportImpactStoreToFS	FGI_ExportImpactStoreToSTA.sh	0 17 * * *	FRBC0001 -> STA
7	fgi.main.ExportCompetitor	FGI_ExportCompetitorToBPM.sh	3 0 1 7 7 - 3 1 * *	BPM06003O
8	fgi.main.ExportImpactStoreFlowToBPM	FGI_ExportImpactStoreToBPM.sh	3 0 1 7 7 - 3 1 * *	BPM06001O SFTP only
8 b	fgi.main.StartK2WorkFlow	Separate scheduler	Confirm with Ops	K2 REST StartInstance
9	fgi.main.ExportOpenStore	FGI_ExportNewStoreToBPM.sh	3 0 1 7 7 - 3 1 * *	BPM06002O
10	fgi.main.NotifyNoReceiveData	FGI_NotifyNoReceiveData.sh	0 07 * * *	Read-only STA ACK watchdog

4. Data Model and Database Dictionary

Type inference SGI V4 ระบุว่า repository ไม่มี DDL ครบทุกตาราง ดังนั้นชนิดข้อมูลบางส่วนถูก infer จาก RowMapper, PreparedStatement และ Oracle functions; ต้องยืนยันกับ DBA ก่อนสร้าง schema ใหม่

4.1 Relationship Map



4.2 Status Domains

Domain	Table	Values / Meaning
FLAG_VERIFY	FGI_IMPACT_STORE / SALES	W Wait; P On Process; Y Accept; N Deny; Z wait-pay filler
FLAG_ACTION	ON_PROCESS	Y Active; W Wait/closing; N Ended; active = Y,W
FLAG_GEN_FLOW	IMPACT_STORE	W waiting K2; Y generated/skip; N ineligible
COMPENSATE_STATUS	COMPENSATE	I Init; C Closed; A Approved; N Rejected; S Final; Z wait slot -> output S

Domain	Table	Values / Meaning
IMPACT_TYPE	STORE_INFO	01 default; 02 60 working days/open after threshold; 03 working day !=60; 04 STA-sourced
DATA_NAME	CONFIRM_RECEIVE_DATA	Discriminator for IMPACT_STORE, NEW_STORE, IMPACT_COMPETITOR, COMPENSATE_*_I/N, SALES
RETURN_CODE	CONFIRM_RECEIVE_DATA	NULL = no ACK; W = received
DATASOURCE	Multiple	ALM = ALLMAP; STA = Franchise Statement; HRS = HR feed

4.3 Sequence Catalog

Sequence	Target
SEQ_FGI_IMPACT_STORE	FGI_IMPACT_STORE.IMPACT_STORE_ID
SEQ_FGI_IMPACT_PROCESS	FGI_IMPACT_STORE_ON_PROCESS.IMPACT_PROCESS_ID
SEQ_FGI_COMPENSATE_I	FGI_IMPACT_STORE_COMPENSATE.COMPENSATE_I_ID
SEQ_FGI_COMPENSATE_N	FGI_NEW_STORE_COMPENSATE.COMPENSATE_N_ID
SEQ_FGI_STORE_INFO_I	FGI_IMPACT_STORE_INFO.IMPACT_STORE_INFO_ID
SEQ_FGI_STORE_INFO_N	FGI_NEW_STORE_INFO.NEW_STORE_INFO_ID
SEQ_FGI_IMPACT_COMPETITOR	FGI_IMPACT_COMPETITOR.IMPACT_COMPETITOR_ID

4.4 Core Table Dictionary

FGI_IMPACT_STORE

Column / Group	Type	Null	Key	Notes
IMPACT_STORE_ID	NUMBER	N	PK	SEQ_FGI_IMPACT_STORE
MONTH / YEAR	NUMBER	N	Period	Business period
STORECODE_I	VARCHAR2	N	FK	Business key across FGI tables
STORECODE_N	VARCHAR2	Y	FK	New store
BRANCHTYPE_I / _N	VARCHAR2	Y	-	Eligibility / disqualifier
OPENDATE_I / _N	DATE	Y	-	Store open dates
JURISTIC_* / SBP_*	Mixed	Y	-	Enrichment from franchise contract
GROWTH_RATE_DIFF	NUMBER	Y	-	K2 gate <= -10
SALES_STATUS	VARCHAR2	Y	-	K2 gate Y/N
FLAG_VERIFY	VARCHAR2	N	-	W/P/Y/N
FLAG_GEN_FLOW	VARCHAR2	Y	-	W/Y/N
CREATE_BY / UPDATE_BY	VARCHAR2	Y	-	ALM/STA

FGI_IMPACT_STORE_ON_PROCESS

Column / Group	Type	Null	Key	Notes
IMPACT_PROCESS_ID	NUMBER	N	PK	Hub key
STORECODE_I	VARCHAR2	N	FK	Links impact store
LAST_COMPENSATE_SEQ / _NO	NUMBER	N	-	Period counter

Column / Group	Type	Null	Key	Notes
START/LAST/END MONTH/YEAR	NUMBER	Y	-	Process period range
FLAG_ACTION	VARCHAR2	N	-	Y/W/N
DATASOURCE	VARCHAR2	N	-	ALM/STA
UPDATE_DATE	DATE	Y	-	Stale wait >30 days logic

FGI_IMPACT_STORE_COMPENSATE

Column / Group	Type	Null	Key	Notes
COMPENSATE_I_ID	NUMBER	N	PK	SEQ_FGI_COMPENSATE_I
IMPACT_PROCESS_ID	NUMBER	N	FK	Hub
STORECODE_I	VARCHAR2	N	FK	Impact store
COMPENSATE_SEQ / _NO	NUMBER	N	-	Ordering
COMPENSATE_MONTH / YEAR	NUMBER	N	-	Period
FORECAST / ADJUST	NUMBER	Y	-	Compensation value
COMPENSATE_STATUS	VARCHAR2	N	-	I/C/A/N/S/Z
STMT_MONTH / YEAR	NUMBER	Y	-	Statement period
COMMENT / audit dates	Mixed	Y	-	Operational metadata

FGI_IMPACT_STORE_INFO

Column / Group	Type	Null	Key	Notes
IMPACT_STORE_INFO_ID	NUMBER	N	PK	SEQ_FGI_STORE_INFO_I
IMPACT_PROCESS_ID	NUMBER	N	FK	Hub
STORECODE_I	VARCHAR2	N	FK	Impact store
COMPENSATE period/seq	NUMBER	N	-	Snapshot period
BPM_COMPENSATE_SEQ / _NO	NUMBER	Y	-	Assigned for BPM
Store/juristic/franchise fields	Mixed	Y	-	Snapshot
DV/GM/AVP + SBP approver fields	VARCHAR2	Y	-	Export to BPM
IMPACT_TYPE	VARCHAR2	Y	-	0 1 - 0 4
URL_ALL_MAP	VARCHAR2	Y	-	Encoded parameters

FGI_NEW_STORE_INFO / COMPENSATE

Column / Group	Type	Null	Key	Notes
NEW_STORE_INFO_ID / COMPENSATE_N_ID	NUMBER	N	PK	Sequences N
IMPACT_PROCESS_ID	NUMBER	N	FK	Hub
IMPACT_STORE_ID	NUMBER	Y	FK	Impact pair
STORECODE_N / STORECODE_I	VARCHAR2	-	FK	New and impact store
COMPENSATE period/seq	NUMBER	N	-	Period
FORECAST_N / ADJUST_N	NUMBER	Y	-	Value
FORECAST_PERCENT_N / ADJUST_PERCENT_N	NUMBER	Y	-	Percent
Store/juristic/franchise	Mixed	Y	-	Snapshot

Column / Group	Type	Null	Key	Notes
fields				

FGI_IMPACT_STORE_SALES / SALES_TRN

Column / Group	Type	Null	Key	Notes
STORECODE_I + MONTH + YEAR	Mixed	N	Business key	Sales header
OPENDATE_I / _N	DATE	Y	-	Window anchor
GROWTH_RATE_BEFORE/ AFTER/DIFF	NUMBER	Y	-	Computed result
TOTAL_WORKING_DAY	NUMBER	Y	-	Raw record count
FLAG_VERIFY	VARCHAR2	N	-	W/P/Y/N
TRN.SEQ	NUMBER	N	-	Day offset
TRN 4 date/sales windows	Mixed	Y	-	Before/after, past/present
TRN DIFF fields	NUMBER	Y	-	Outlier threshold 50
IMPORT_FILE	VARCHAR2	Y	-	Source file

FGI_IMPACT_COMPETITOR

Column / Group	Type	Null	Key	Notes
IMPACT_COMPETITOR_ID	NUMBER	N	PK	SEQ_FGI_IMPACT_COMPETITOR
STORECODE_I	VARCHAR2	N	FK	Impact store
COMPET_ID	VARCHAR2	Y	-	Competitor ID
NAME_TH	VARCHAR2	Y	-	Source NAMT
NAME	VARCHAR2	Y	-	English
BRANCH_TH	VARCHAR2	Y	-	Source BRANCHT
ZONE/SUBZONE	VARCHAR2	Y	-	Location
OPEN/CLOSE_DATE	DATE	Y	-	Dates
MONTH/YEAR	NUMBER	N	-	Source period
DATASOURCE	VARCHAR2	N	-	ALM

FGI_CONFIRM_RECEIVE_DATA

Column / Group	Type	Null	Key	Notes
DATA_NAME	VARCHAR2	N	Discriminator	Object type
TRANSACTION_PK	NUMBER	N	Poly FK	Target PK depends on DATA_NAME
MONTH / YEAR	NUMBER	N	-	Period
INTERFACE_TYPE	VARCHAR2	Y	-	WS or filename / W by some jobs
RETURN_CODE	VARCHAR2	Y	-	NULL or W
RECEIVE_DATE	DATE	Y	-	Receipt date
CREATE_DATE	DATE	Y	-	Age/purge/watchdog

FCS_QSSI_SCORE / FCS_TMP_QSSI_SCORE

Column / Group	Type	Null	Key	Notes
STORE_ID	VARCHAR2	N	Business key	Store
CATEGORY	NUMBER	N	-	Includes 1,8,9,10,12,16

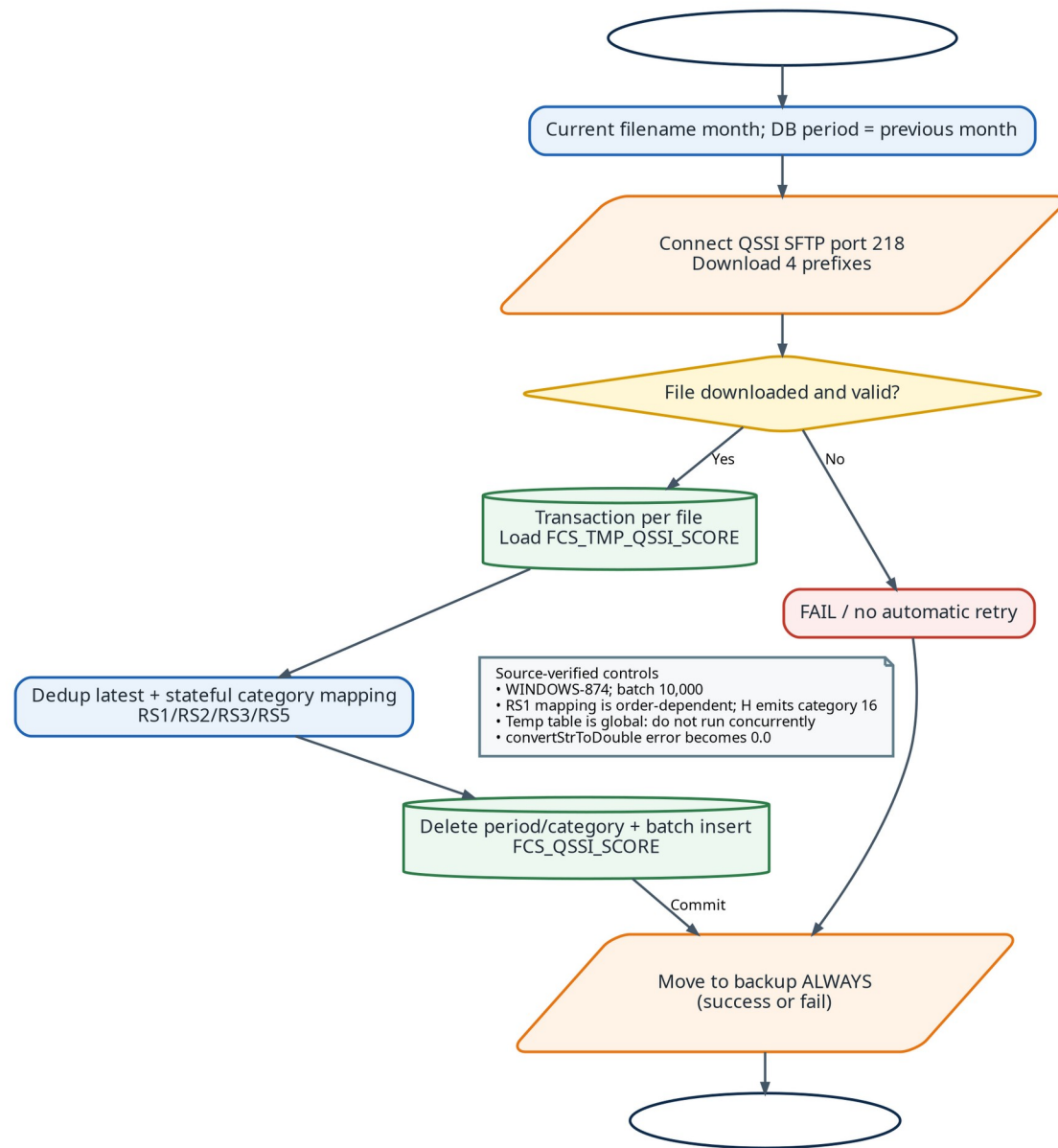
Column / Group	Type	Null	Key	Notes
MONTH / YEAR	String/Number	N	-	Used in to_date(year month)
SCORE	NUMBER	Y	-	NVL to zero during export
TMP DATA_DESC/MAX/PERCENT /FILE/CHECK_DATE	Mixed	Y	-	Dedup and mapping staging

4.5 External Objects and Master Joins

- SQL Server views: allmapssa.SEVEN_IMPACT_VIEW and allmapssa.COMPETITOR_IMPACT_VIEW.
- Oracle views: V_FGI_SBP_APPROVER and V_FGI_IMPACT_USER.
- Master joins: MAS_STORE, MAS_STORE_ORGANIZE, MAS_EMPLOYEE, MAS_PARAM, FR_STORE, JURISTIC, FRANCHISEE, BUSINESS_USER, BUSINESS_USER_GROUP.
- FGI_CONFIRM_RECEIVE_DATA.TRANSACTION_PK is polymorphic; a new schema should prefer typed foreign keys or a discriminated association.

5. Detailed Job Specifications

Job 1 - ImportQSSI		IMPORT
Main Class	fcs.main.ImportQSSI	
Script / Trigger	FCS_ImportQSSI.sh / monthly	
Transaction	Per file TransactionTemplate + savepoint	
Primary Tables	FCS_TMP_QSSI_SCORE, FCS_QSSI_SCORE, COMMON_CODE	



Source-verified control flow

Inputs and Preconditions

- Remote directory /export/qssishare/onl/qssi/textfile/SBP/QSSI_Monthly/
- SFTP port 218; four prefixes mrs1trnf_, mrs2trnf_, mrs3trnf_, mrs5trnf_
- Local input /appshare/SPS/FCS/interface_data/in/; WINDOWS-874; batch size 10,000

Business Rules and Implementation Notes

- Filename uses the current month, while the DB period written is the previous month.
- RS1 mapping is stateful: D+E, F+G+H, I+J, K+L; H also emits category 16.
- Backup move runs even when import fails, so there is no automatic retry.
- Temp table is deleted per file and must not be processed concurrently.

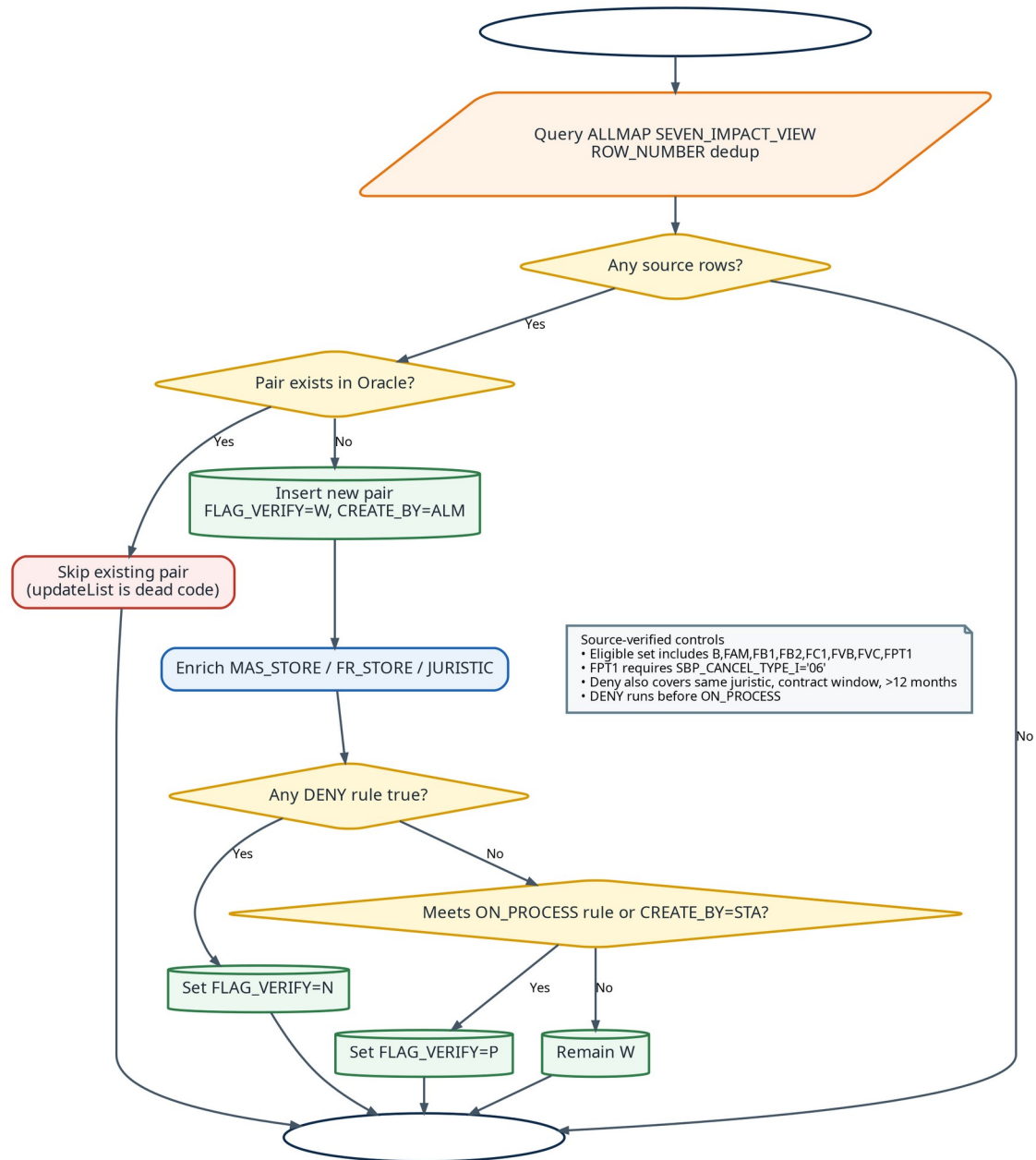
Output, Error Handling and Re-run

- Output FCS_QSSI_SCORE; data later feeds Job 6 QSSI categories 8,9,12,1,10,16.
- On exception, rollback to savepoint and send failure email; source file still moves to backup.

Job 2 - ImportImpactStore

IMPORT + RULES

Main Class	fgi.main.ImportImpactStore
Script / Cron	FGI_ImportImpactStore.sh / 0 07 7 * *
Transaction	One transaction + savepoint
Primary Tables	FGI_IMPACT_STORE + master/franchise joins



Source-verified control flow

Inputs and Preconditions

- Argument ZONES|YYYY|MM or ALL|YYYY|MM; default period follows modifyDateToString behavior.
- Source allmapssa.SEVEN_IMPACT_VIEW; target PK SEQ_FGI_IMPACT_STORE.

Business Rules and Implementation Notes

- Existing pairs are skipped: updateList is dead code.

- Eligible set includes B,FAM,FB1,FB2,FC1,FVB,FVC,FPT1; FPT1 requires cancel type 06.
- DENY covers branch N=F, same juristic, SBP contract not covering period, or record older than 12 months.
- DENY runs before ON_PROCESS; CREATE_BY=STA can become P automatically.

Output, Error Handling and Re-run

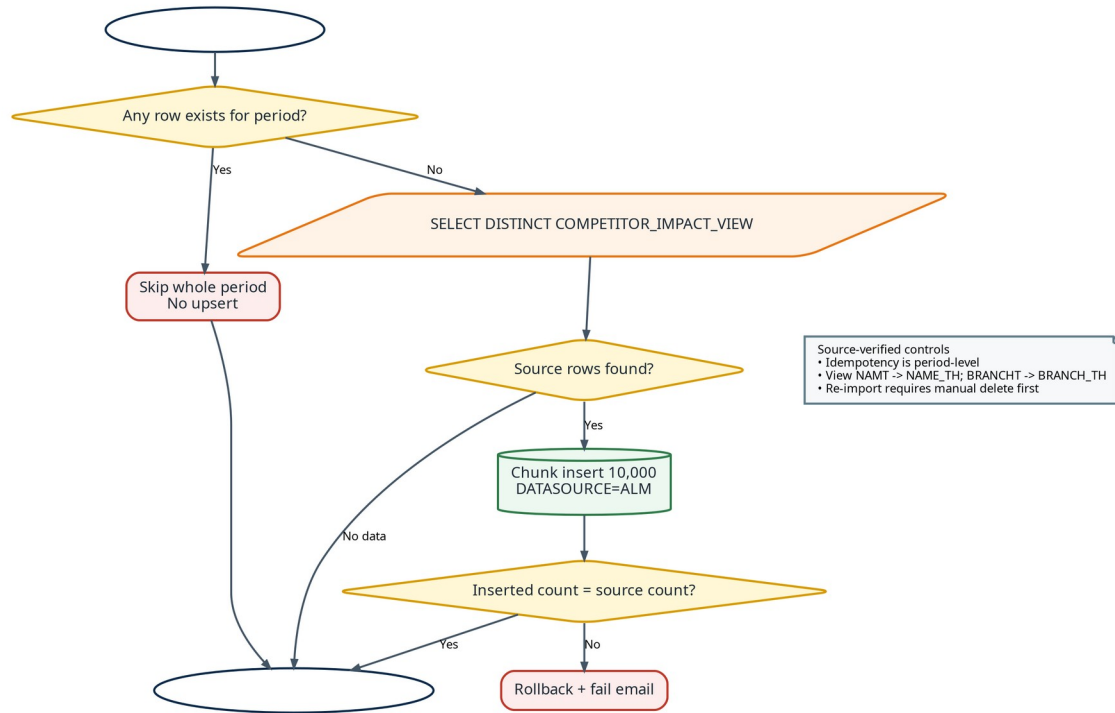
- New rows start W and CREATE_BY=ALM; enrichment comes from MAS_STORE/FR_STORE/JURISTIC.
- Re-run must consider skipped existing pairs and master rows missing from INNER JOIN.

```
DENY set: ('B','FAM','FB1','FB2','FC1','FVB','FVC','FPT1')
FPT1 is eligible only with SBP_CANCEL_TYPE_I='06'
Existing pair -> skip (updateList is dead code)
```

Job 3 - ImportImpactCompetitor

IMPORT

Main Class	fgi.main.ImportImpactCompetitor
Script / Cron	Confirm with scheduler; 0 07 7 * *
Transaction	One transaction + savepoint; chunks of 10,000
Primary Table	FGI_IMPACT_COMPETITOR



Source-verified control flow

Inputs and Preconditions

- Argument YYYY|MM; source COMPETITOR_IMPACT_VIEW.

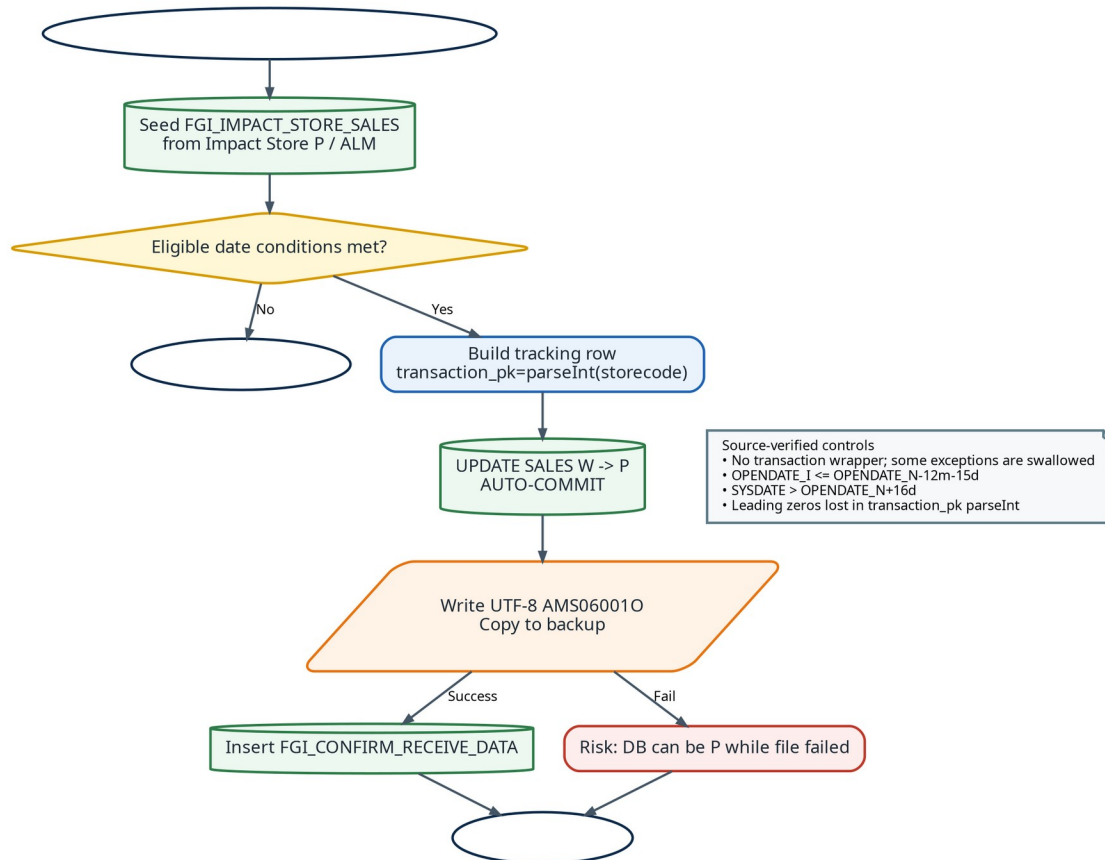
Business Rules and Implementation Notes

- Duplicate guard is period-level; any existing row causes the entire period to be skipped.
- Source column NAMT maps to NAME_TH and BRANCHT maps to BRANCH_TH.
- DATASOURCE is literal ALM.

Output, Error Handling and Re-run

- No upsert. To re-import a period, delete existing rows first and reconcile source count.

Job 4 - PrepareImpactStoreToIAS		PREPARE / EXPORT
Main Class	fgi.main.PrepareImpactStoreToIAS	
Script / Cron	FGI_ExportImpactStoreToAMS.sh / 0 16 7-16 * *	
Transaction	None - auto-commit	
Output	06001	8□



Source-verified control flow

Inputs and Preconditions

- Seeds SALES from Impact Store P/ALM, excludes active ON_PROCESS.
- Eligible when impact store is open at least 12 months + 15 days before new store and current date passes new-store +16 days.

Business Rules and Implementation Notes

- UPDATE W->P happens before file writing and cannot roll back.
- `transaction_pk` uses `parseInt(STORECODE_I)`, which loses leading zeros.
- Backup is copy, not move; live file remains for Job 5.

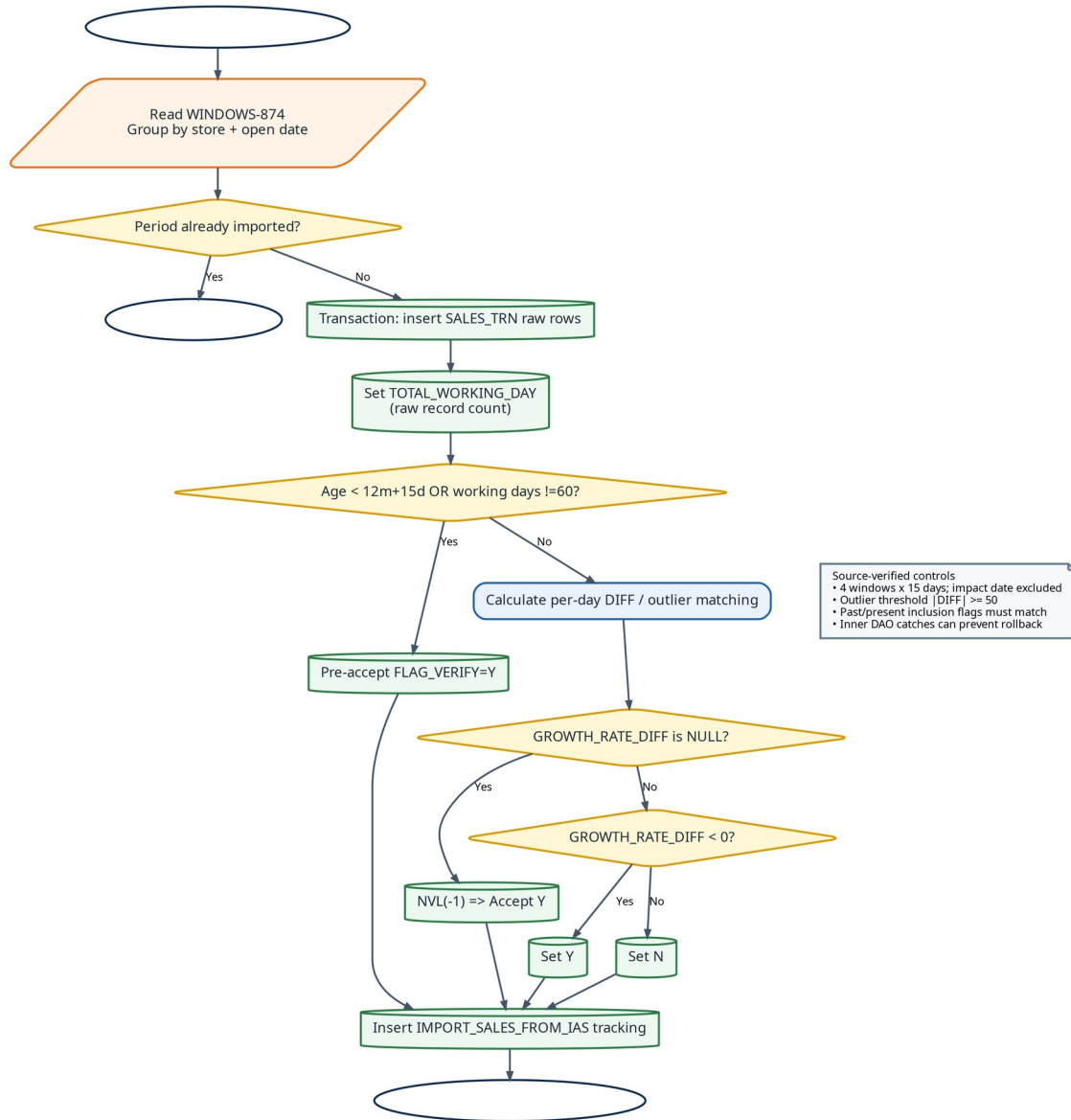
Output, Error Handling and Re-run

- Tracking `DATA_NAME=IMPACT_STORE_SALES` with NULL return code/receive date.
- Recovery must reconcile DB flag, output file, backup and tracking manually.

Job 5 - ImportImpactSaleFromIAS

IMPORT / CALCULATION

Main Class	fgi.main.ImportImpactSaleFromIAS
Script / Cron	FGI_ImportImpactStoreSale.sh / 30 16 7-16 * *
Transaction	One transaction + savepoint per file
Input	AMS06001L_yyyyMMddHHmm.txt, WINDOWS-874



Source-verified control flow

Inputs and Preconditions

- Four 15-day windows around OPENDATE_N; impact date itself is excluded.
- Input fields: STORECODE_I | OPENDATE_N | SALES_DATE | SALES_AMOUNT.

Business Rules and Implementation Notes

- TOTAL_WORKING_DAY is raw record count, including records outside the four windows.
- Pre-accept Y occurs when store age is insufficient or working day count is not 60.
- Outlier threshold is |DIFF| >= 50 and inclusion pairs past/present flags.

- Final rule NVL(GROWTH_RATE_DIFF,-1)<0 means NULL becomes Y.

Output, Error Handling and Re-run

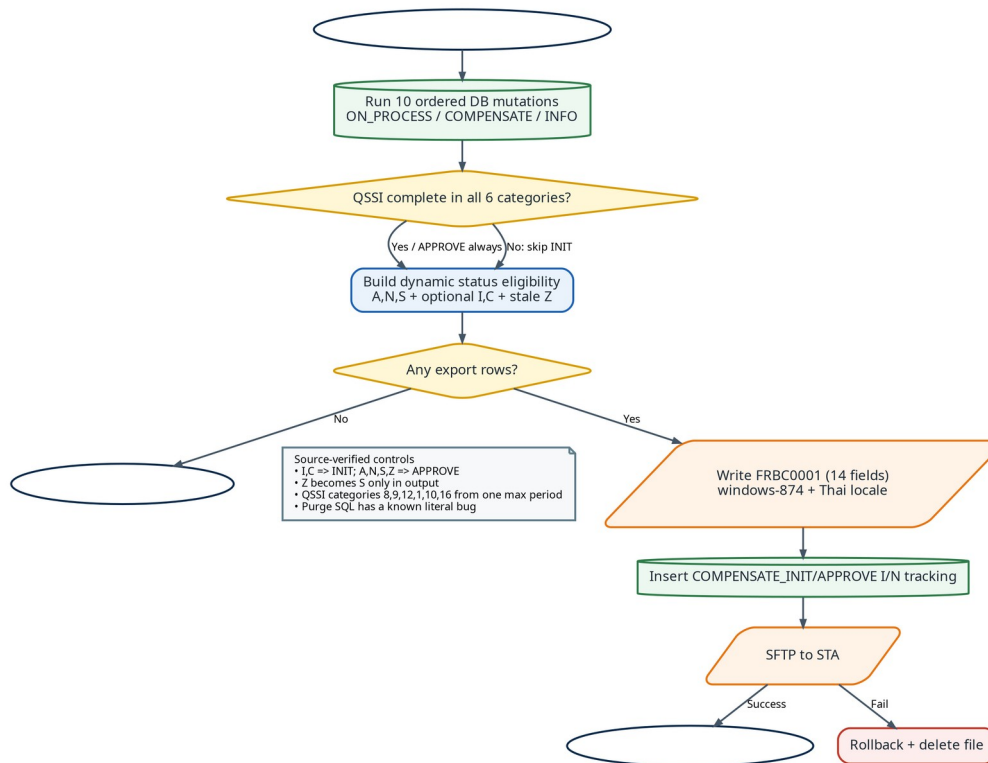
- Updates SALES_TRN and SALES header, inserts IMPORT_SALES_FROM_IAS tracking, backs up input.
- Unit tests must cover zero-sales / missing data, leap-Feb windows and paired outlier behavior.

```
PRE-ACCEPT: OPENDATE_I > ADD_MONTHS(OPENDATE_N,-12)-15 OR TOTAL_WORKING_DAY != 60 -> Y  
FINAL: CASE WHEN NVL(GROWTH_RATE_DIFF,-1) < 0 THEN 'Y' ELSE 'N' END
```

Job 6 - ExportImpactStoreToFS / STA

STATE SYNC + EXPORT

Main Class	fgi.main.ExportImpactStoreToFS
Script / Cron	FGI_ExportImpactStoreToSTA.sh / 0 17 * * *
Transaction	One TransactionTemplate + savepoint covering sync/file/tracking/SFTP
Output	FRBC0001_yyyyMMddHHmmss.txt, windows-874, 14 fields



Source-verified control flow

Inputs and Preconditions

- Runtime: dateStartInitToSTA=7; numWaitPay=3; QSSI categories 8,9,12,1,10,16.
- Runs 10 ordered mutations across ON_PROCESS, COMPENSATE, STORE_INFO and NEW_STORE tables before export.

Business Rules and Implementation Notes

- Dynamic eligible status list: A,N,S; add I,C when day>=7 and complete QSSI; include stale Z branch.
- I,C map to COMPENSATE_INIT_I/N; A,N,S,Z map to COMPENSATE_APPROVE_I/N.
- Z remains Z in DB and is converted to S only in output.
- QSSI must be complete across all six categories for INIT path and comes from one max period in the 3-month window.
- FRBC0001 date/period fields use Thai/Buddhist locale.

Output, Error Handling and Re-run

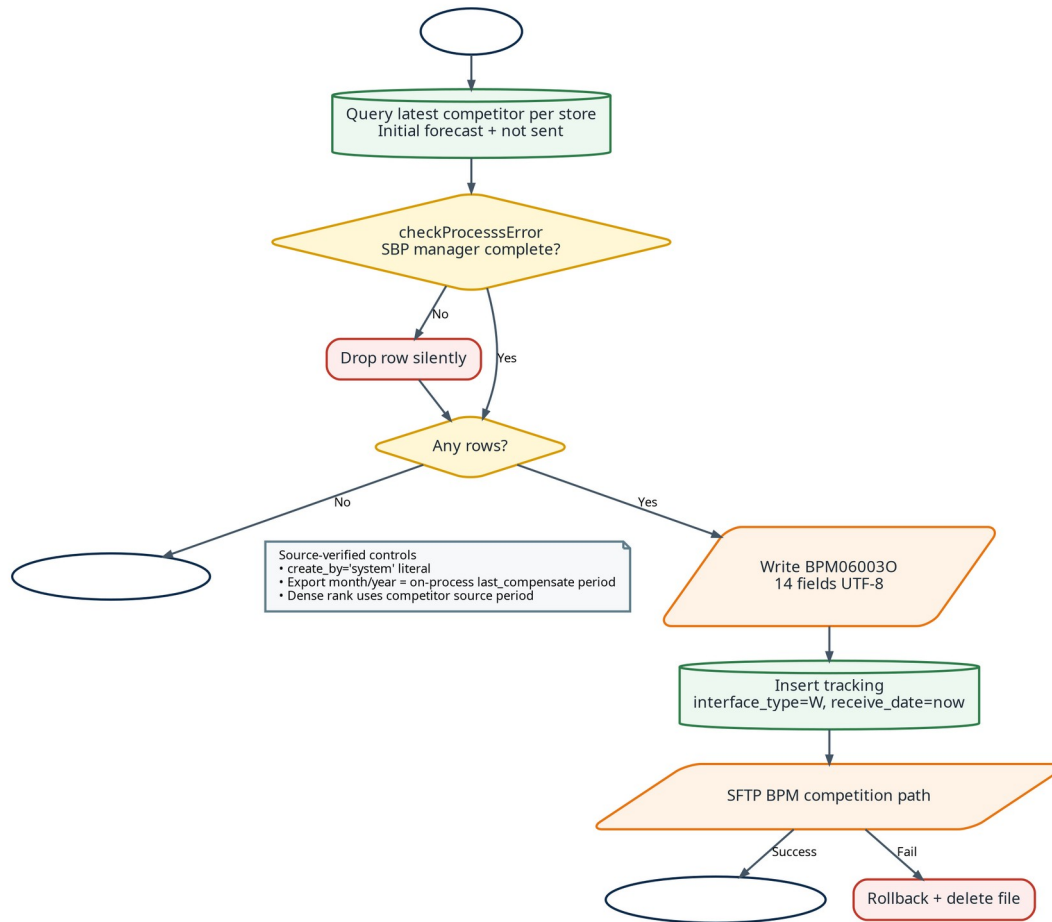
- SFTP failure rolls back and deletes file; success moves backup.
- Known source bug: purge SQL treats two data names as one literal string, so compensate tracking is not purged.
- A single invalid date can cause mapData... to return null and cancel the entire file.

STATUS: A,N,S; add I,C when date gate + QSSI complete; include stale Z
 DATA_NAME: I,C -> INIT_* | A,N,S,Z -> APPROVE_*; output maps Z -> S

Job 7 - ExportCompetitor

EXPORT BPM

Main Class	fgi.main.ExportCompetitor
Script / Cron	FGI_ExportCompetitorToBPM.sh / 30 17 7-31 * *
Transaction	One transaction + savepoint
Output	BPM06003O_YYYYMMDDHHmm.txt, UTF-8, 14 fields

*Source-verified control flow*

Inputs and Preconditions

- Selects latest competitor period per store in active ALM process with initial compensation forecast.

Business Rules and Implementation Notes

- create_by is literal 'system'.
- Export month/year come from ON_PROCESS last_compensate period, not competitor source period.
- Rows with incomplete SBP manager data are silently removed.

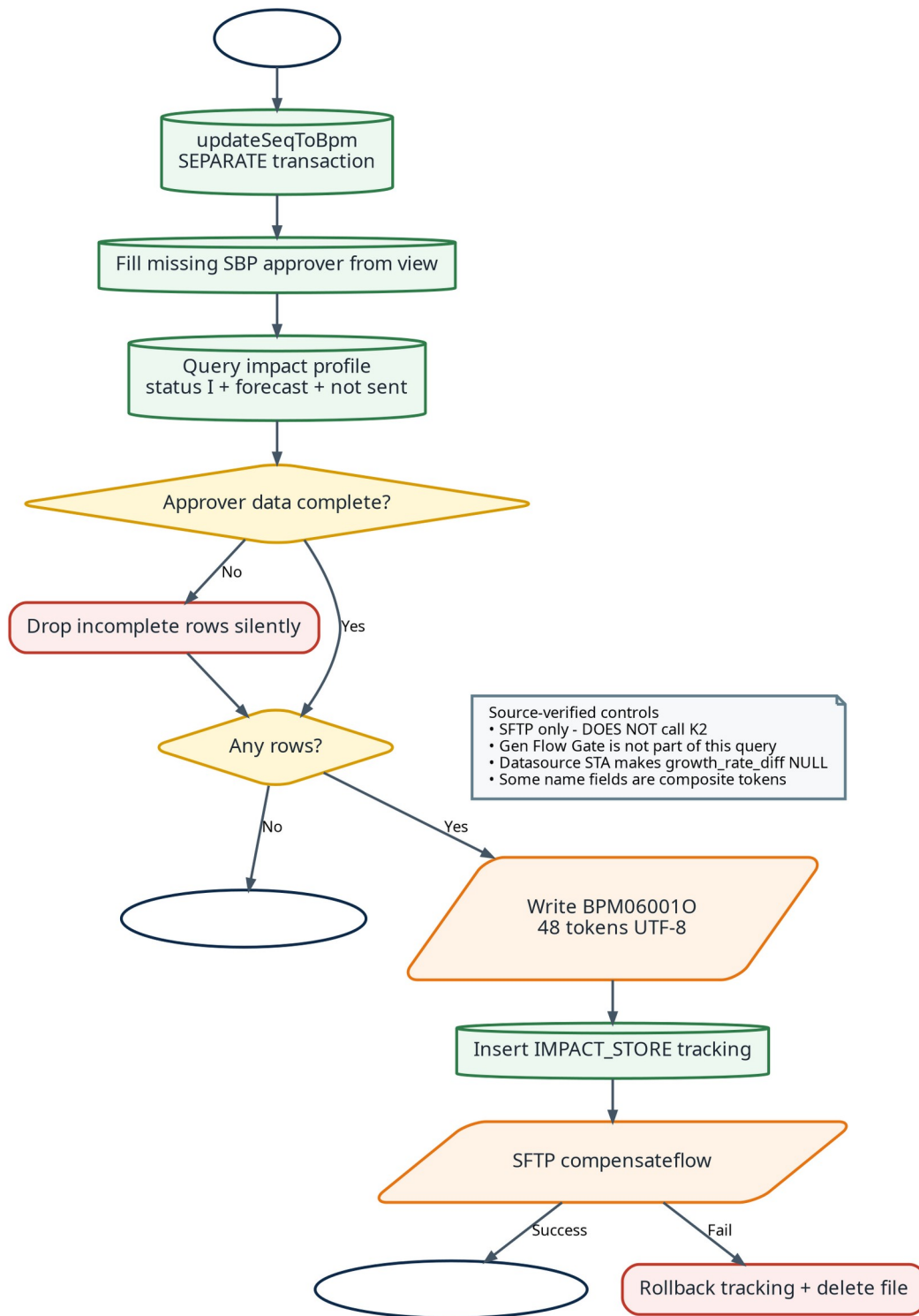
Output, Error Handling and Re-run

- Tracking DATA_NAME=IMPACT_COMPETITOR; interface_type=W; receive_date=now; SFTP competition path.

Job 8 - ExportImpactStoreFlowToBPM

EXPORT BPM - SFTP ONLY

Main Class	fgi.main.ExportImpactStoreFlowToBPM
Script / Cron	FGI_ExportImpactStoreToBPM.sh / 30 17 7-31 * *
Transaction	updateSeqToBpm separate transaction + main transaction
Output	BPM06001O_YYYYMMddHHmm.txt, UTF-8, 48 tokens



Source-verified control flow

Inputs and Preconditions

- Exports impact profile, compensation and SBP approver data to BPM/FGI/Inbound/compensateflow/.

Business Rules and Implementation Notes

- This job does not call K2. Gen Flow Gate belongs only to Job 8b.
- growth_rate_diff is NULL for DATASOURCE=STA.
- Several name fields are composite first-name + space + last-name tokens.
- Rows with incomplete approvers are removed by checkProcessError.

Output, Error Handling and Re-run

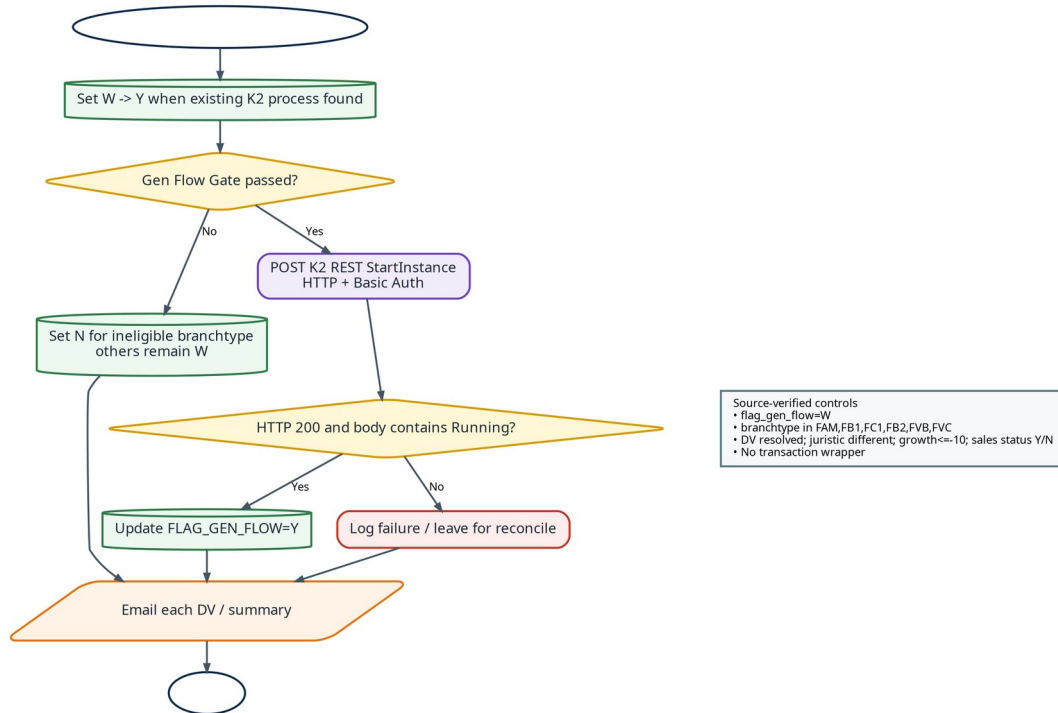
- Tracking DATA_NAME=IMPACT_STORE, transaction_pk=IMPACT_STORE_INFO_ID, return_code=W, receive_date=now.
- SFTP fail rolls back tracking and deletes file.

Separation of Concerns Job 8 creates and uploads BPM06001O only. It must not contain K2 StartInstance logic. Job 8b owns the K2 gate, REST call and FLAG_GEN_FLOW reconciliation.

Job 8b - StartK2WorkFlow

K2 REST

Main Class	fgi.main.StartK2WorkFlow
Trigger	Separate scheduler - confirm with Operations
Transaction	No transaction wrapper; flag updates auto-commit
Endpoint	K2 REST StartInstance over HTTP Basic Auth



Source-verified control flow

Inputs and Preconditions

- Gate requires FLAG_GEN_FLOW=W, branch type FAM/FB1/FC1/FB2/FVB/FVC, resolved DV, different juristic, growth_rate_diff<=-10 and SALES_STATUS in Y,N.

Business Rules and Implementation Notes

- updateBeforeSelect marks rows with an existing impact process as Y to prevent duplicate starts.
- Success requires HTTP 200 and response body containing Running.
- Successful rows are updated to Y; ineligible branch types are set N; failures require reconciliation through FLAG_GEN_FLOW.

Output, Error Handling and Re-run

- Sends emails grouped by OPT_DV_USER_ID.
- Current state uses HTTP and hardcoded Basic Auth; move to HTTPS and secret manager during re-implementation.

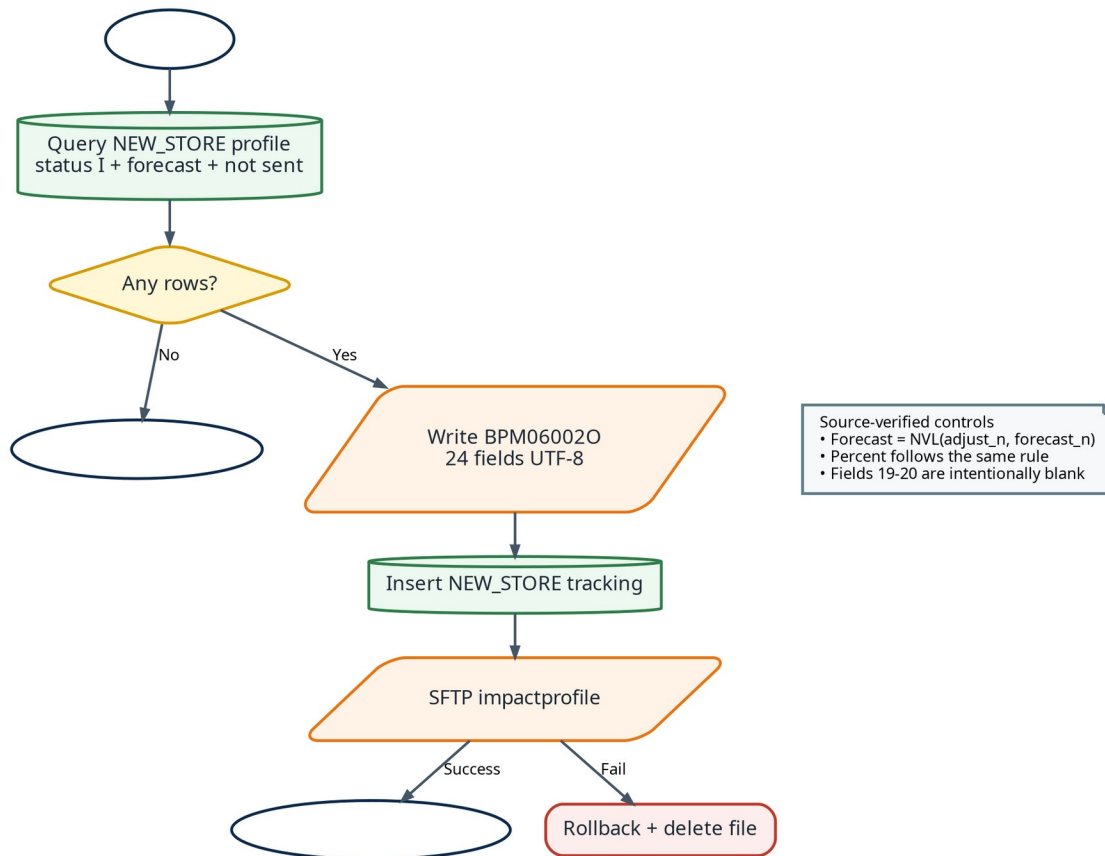
```

FLAG_GEN_FLOW='W'
BRANCHTYPE_I IN ('FAM','FB1','FC1','FB2','FVB','FVC')
OPT_DV_USER_ID IS NOT NULL
JURISTIC_N <> JURISTIC_I
GROWTH_RATE_DIFF <= -10
SALES_STATUS IN ('Y','N')
  
```

Job 9 - ExportOpenStore

EXPORT BPM

Main Class	fgi.main.ExportOpenStore
Script / Cron	FGI_ExportNewStoreToBPM.sh / 30 17 7-31 * *
Transaction	One transaction + savepoint
Output	BPM06002O_YYYYMMDDHHmm.txt, UTF-8, 24 fields



Source-verified control flow

Inputs and Preconditions

- Exports new-store profile and forecast to BPM/FGI/Inbound/impactprofile/.

Business Rules and Implementation Notes

- Forecast and percent use NVL(adjust, forecast).
- Fields 19-20 (adjust values) are intentionally blank in the file.
- distance_unit is selected but not emitted.

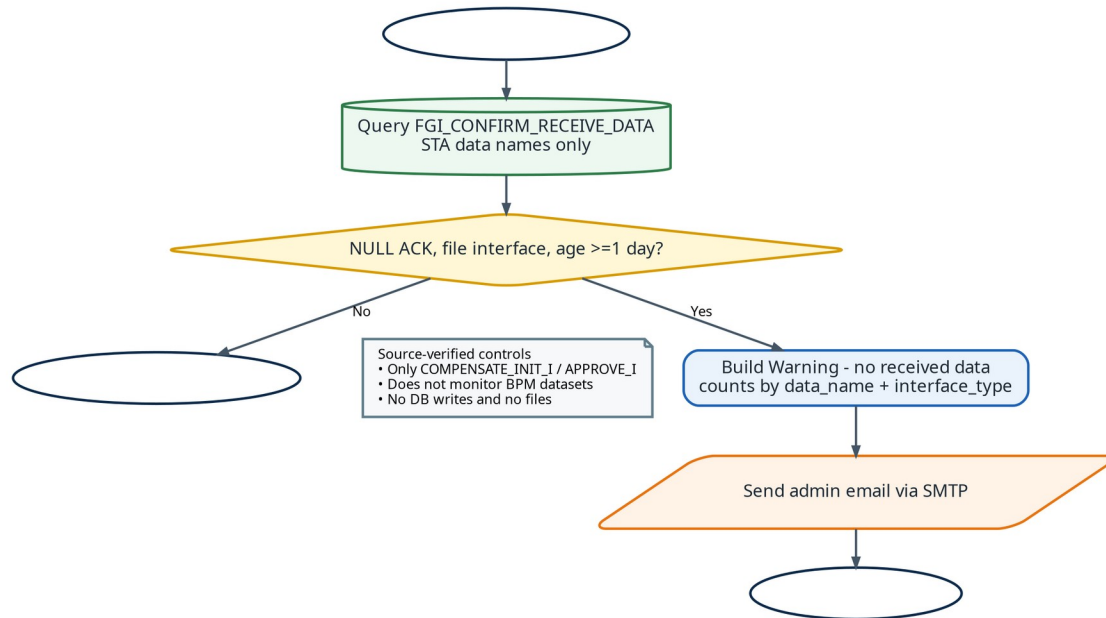
Output, Error Handling and Re-run

- Tracking DATA_NAME=NEW_STORE; SFTP success moves backup; failure rolls back and deletes file.

Job 10 - NotifyNoReceiveData

WATCHDOG

Main Class	fgi.main.NotifyNoReceiveData
Script / Cron	FGI_NotifyNoReceiveData.sh / 0 07 * * *
Transaction	Read-only
Output	Warning email only

*Source-verified control flow*

Inputs and Preconditions

- Queries FGI_CONFIRM_RECEIVE_DATA for COMPENSATE_INIT_I and COMPENSATE_APPROVE_I.

Business Rules and Implementation Notes

- Pending means RETURN_CODE IS NULL, INTERFACE_TYPE != WS and CREATE_DATE <= SYSDATE-1.
- Aggregation includes ACKed rows; count_data - count_return_code is pending count.
- Does not monitor BPM datasets.

Output, Error Handling and Re-run

- No DB writes and no files; re-run is safe but avoid duplicate warnings.

6. Cross-Cutting Concerns

6.1 Encoding and File Contracts

Interface	Direction	Encoding	Owner
QSSI mrs*	Inbound	WINDOWS-874	Job 1
AMS06001O	Outbound	UTF-8	Job 4
AMS06001I	Inbound	WINDOWS-874	Job 5
FRBC0001	Outbound	windows-874 + Thai locale	Job 6
BPM06001O	Outbound	UTF-8	Job 8
BPM06002O	Outbound	UTF-8	Job 9
BPM06003O	Outbound	UTF-8	Job 7
Email	Outbound	TIS-620	All jobs

6.2 Credentials and Security

Current State SFTP credentials are stored in plaintext configuration; K2 Basic Auth is hardcoded; StrictHostKeyChecking=no; K2 endpoint uses HTTP. These are facts of the current implementation, not recommended target design.

System	Endpoint	User	Current Storage
QSSI SFTP	sezfssa02-office.cpall.co.th:218	qssi	Plaintext config; masked in document
STA SFTP	sestabr01.cpall.co.th:22	staapp	Plaintext config; masked
BPM SFTP	seeaiff07.cpall.co.th:22	vdeu	Plaintext config; masked
K2 REST	sesomapua01.7eleven.cp.co.th:81	7ELEVEN\bpmk2_fcs	Hardcoded Basic Auth; masked

- Target state: externalize secrets to Vault/Secret Manager, enforce TLS/HTTPS, manage known_hosts and mask credentials/payloads in logs.

6.3 Transaction and Re-run Matrix

Entry	Pattern	Operational Note
1	Per file transaction	Atomic DB; backup even on fail
2	One transaction	Insert/enrich/deny/onprocess
3	One transaction	Chunk insert + count reconcile
4	No transaction	Highest DB/file inconsistency risk
5	Per file transaction	Inner catches may not rollback
6	One transaction	Sync + file + tracking + SFTP
7	One transaction	Rollback/delete on fail
8	Two transactions	Sequence assignment separate
8 b	No transaction	Reconcile with FLAG_GEN_FLOW
9	One transaction	Rollback/delete on fail
10	Read-only	Safe to rerun; duplicate email risk

6.4 Environment and Datasources

- Oracle datasource uses SingleConnectionDataSource, not a connection pool; BaseJdbc.jdbcTemplate is static and assumes single-run execution.
- SQL Server datasource is GSMALLMAP with allmapssa in production and sde in QA.
- The committed jdbc.properties referenced by SGI V4 points to STQA test; verify active environment before every production run.
- Run from project root so config.xml and relative classpath resources resolve correctly: `java -cp "frFCSJar.jar:lib/*:." <mainClass> [args]`.

6.5 Mail Routing

Job	Routing
1	storeretention (Cc blank)
2	go-sbp hardcoded; template 34
3	config mailTo / storeretention
4,5	go-sbp via shared helper
6,7,8,9	storeretention + mailToBPM when file generated
8b	Per DV/GM user
10	storeretention warning

7. Operational Runbook

7.1 Pre-Run Checklist

- Confirm business date, scheduler window and active datasource/environment.
- Confirm upstream file or ALLMAP view is available and has expected period.
- Check previous run status, live output folder, backup folder and FGI_CONFIRM_RECEIVE_DATA.
- Confirm no concurrent Job 1 run because FCS_TMP_QSSL_SCORE is shared.
- Verify disk space, SFTP reachability and SMTP route.
- For Job 8b, confirm K2 endpoint and scheduler ownership separately from Job 8.

7.2 Post-Run Checklist

- Validate row counts and state flags against source records.
- Verify file name, field count, encoding, locale and remote destination.
- Verify tracking DATA_NAME / TRANSACTION_PK / month / year and ACK status.
- Review warning/failure mail and log path; confirm backup move/copy semantics.
- For Job 6, reconcile all 10 state mutations before confirming successful business output.

7.3 Re-run Decision Guide

Job	Constraint	Re-run Action
1	File moved even on failure	Restore source file only after checking target period and temp table
2	Existing pair skipped	Delete/repair pair deliberately; rerun will not update
3	Period guard	Delete period before re-import
4	Auto-commit W->P	Reconcile SALES flags, file and tracking before rerun
5	Period guard + raw count	Delete/repair SALES_TRN carefully; recalc header
6	Transaction normally protects	Check known purge bug and remote overwrite semantics
7 - 9	Tracking prevents duplicate	Check backup + tracking + remote file
8b	Auto-commit flags	Reset FLAG_GEN_FLOW only after K2 instance check
10	Read-only	Safe; may resend warning

7.4 Monitoring Metrics

Area	Recommended Metrics
Scheduler	Run start/end, duration, status per entry point
Input	Files expected/found, source row count, invalid lines
Database	Rows inserted/updated/skipped, W/P/Y/N counts, compensation status counts
Interfaces	File size, row count, SFTP result, remote path, backup result
Tracking	Outstanding ACK age, duplicate prevention count, pending by DATA_NAME
K2	Gate candidates, HTTP Running, failed starts, FLAG_GEN_FLOW W/Y/N
Quality	QSSI six-category completeness, sales NULL diff, working day !=60
Alerts	No ACK >1 day, environment mismatch, credentials/TLS finding

8. Interface and Control Catalog

Interface	Direction	Encoding	Fields	Purpose
mrs1/2/3/5tmf_yyyyMM	QSSI -> Job 1	WINDOWS-874	Variable	QSSI categories
AMS06001O_yyyyMMddH Hmm.txt	Job 4 -> IAS	UTF-8	1 logical record	storecode_i opendate_n
AMS06001I_yyyyMMddH Hmm.txt	IAS -> Job 5	WINDOWS-874	4	Sales daily response
FRBC0001_yyyyMMddHH mmss.txt	Job 6 -> STA	windows-874	14	Compensation + QSSI
BPM06003O_yyyyMMddHHm m.txt	Job 7 -> BPM	UTF-8	14	Competitor
BPM06001O_yyyyMMddHHm m.txt	Job 8 -> BPM	UTF-8	48	Impact profile
K2 StartInstance	Job 8b -> K2	XML/HTTP	Data fields	FIA workflow
BPM06002O_yyyyMMddHHm m.txt	Job 9 -> BPM	UTF-8	24	New store

8.1 Technical Risks and Recommendations

Priority	Risk	Impact	Recommendation
P0	Plaintext/hardcoded credentials + HTTP K2	Credential exposure / MITM	Vault/Secret Manager, TLS, known_hosts
P0	Job 4 auto-commit before file	Lost request or inconsistent flag	Transactional outbox or explicit recovery reconciliation
P0	Job 8/8b separation not understood	Duplicate/missing workflows	Keep decoupling; monitor FLAG_GEN_FLOW and K2 instances
P1	Job 5 NULL diff auto-accept	Incorrect business acceptance	Business sign-off + regression tests
P1	Job 2 existing pair skipped	Master changes not reflected	Confirm intentional behavior or implement controlled update
P1	Job 6 purge literal bug	Tracking table growth	Fix deliberately with data migration and tests
P1	Polymorphic TRANSACTION_PK	Weak referential integrity	Typed keys/discriminated association in target schema
P2	Static JdbcTemplate / shared temp	Concurrency risk	Single-run guard or refactor to scoped datasource/temp
P2	Mixed locale and encoding	Corrupt Thai or date fields	Golden-file tests for each interface

8.2 Re-implementation Checklist

- Create schema from the Data Dictionary and enforce status domains as enums/check constraints.
- Obtain real definitions for SEVEN_IMPACT_VIEW, COMPETITOR_IMPACT_VIEW, V_FGL_SBP_APPROVER and V_FGL_IMPACT_USER.
- Preserve literal thresholds and unusual behavior unless business approves a change: 218, -10, 50%, 60, 12 months, 15 days, 24 months, QSSI 8/9/12/1/10/16, NUM_WAIT_PAY=3.
- Build golden-file tests for all interfaces, encodings, field counts, Thai/Buddhist dates and composite name tokens.
- Write unit tests for every gotcha, especially Job 1 off-by-one, Job 2 dead update, Job 4 auto-commit, Job 5 NULL diff, Job 6 purge bug and Job 8/8b separation.
- Document any intentional bug fix as a behavior change, with migration and rollback plan.

9. Glossary

Term	Meaning
ACK	Acknowledgement from downstream system
ALM	ALLMAP datasource
BPM	Business Process Management file consumer
FCS	Franchise Compensation System module
FGI	Franchise Growth Impact module
IAS/MIS	Sales calculation file interface
2	Workflow engine; started only by Job 8b
STA	Franchise Statement system
QSSI	Quality/Safety score source
W/P/Y/N	Wait / On Process / Accept / Deny

Source Reference This improved document was updated from FGI_FCS_Batch_Job_Technical_Document_Improved_v3.0 using SGI V4 - Source-Verified Implementation Edition. Exact SQL should still be checked against the repository before production re-implementation.